

Arjuna JEE 2024

Maths

DPP: 7

Trigonometric Identities

- Q1** If $\frac{\sin 3}{\sin 5} = k$, then ' k ' is
 (A) $k > 0$ (B) $k < 0$
 (C) $k = 1$ (D) none of these
- Q2** The value of $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$ is
 (A) $\frac{1}{36}$
 (B) $\frac{1}{32}$
 (C) $\frac{1}{18}$
 (D) $\frac{1}{16}$
- Q3** Which of the following is/are correct, when $n \in I$
 (A) $\tan(2n+1)\pi/2$ is not defined
 (B) $\sec(2n+1)\pi/2$ is not defined
 (C) $\operatorname{cosec} n\pi$ is not defined
 (D) $\cot n\pi$ is not defined
- Q4** Which one of the following is true?
 (A) $\sin 1 < \sin 2 < \sin 3$
 (B) $\sin 1 < \sin 3 < \sin 2$
 (C) $\sin 3 < \sin 1 < \sin 2$
 (D) $\sin 3 < \sin 2 < \sin 1$
- Q5** Which of the following is/are true?
 (A) $\sin 3 > \sin 4$
 (B) $\cos 4 > \cos 5$
 (C) $\tan 4 < \tan\left(\frac{9}{2}\right)$
 (D) $\cot 2 > \cot 3$
- Q6** Which of the following is / are correct, when $n \in I$?
 (A) $\tan\left[(2n+1)\frac{\pi}{2}\right]$ is not defined
 (B) $\sec\left[(2n+1)\frac{\pi}{2}\right]$ is not defined
 (C) $\operatorname{cosec} n\pi$ is not defined
 (D) $\cot n\pi$ is not defined
- Q7** The value of $\tan 10^\circ \cdot \tan 50^\circ \cdot \tan 70^\circ$ is
 (A) $\sqrt{3}$
 (B) $\frac{1}{\sqrt{3}}$
 (C) 1
 (D) -1
- Q8** For $A = 133^\circ$, $2 \cos \frac{A}{2}$ is equal to
 (A) $-\sqrt{1+\sin A} - \sqrt{1-\sin A}$
 (B) $-\sqrt{1+\sin A} + \sqrt{1-\sin A}$
 (C) $\sqrt{1+\sin A} - \sqrt{1-\sin A}$
 (D) $\sqrt{1+\sin A} + \sqrt{1-\sin A}$
- Q9** If $0 < \theta < \frac{\pi}{2}$, and if $\frac{y+1}{1-y} = \sqrt{\frac{1+\sin \theta}{1-\sin \theta}}$, then y is equal to
 (A) $\cot \theta/2$
 (B) $\tan \theta/2$
 (C) $\cot \theta/2 + \tan \theta/2$
 (D) $\cot \theta/2 - \tan \theta/2$
- Q10** If $A + B + C = \pi$, then $\tan A + \tan B + \tan C$ is equal to
 (A) $\tan(A+B+C)$
 (B) $\tan A \tan B \tan C$
 (C) $2 \tan A \tan B \tan C$
 (D) $2 \tan(A+B+C)$
- Q11** If $A + B + C = \pi$, then $\cot B \cot C + \cot C \cot A + \cot A \cot B$ is equal to
 (A) 0 (B) -1
 (C) 1 (D) 2



- Q12** If $A + B + C = \pi$, then
 $\sin 2A - \sin 2B + \sin 2C = P$
 $\cos A \sin B \cos C$, then value of P is
 (A) 1 (B) 2
 (C) 3 (D) 4

- Q13** If $A + B + C = 180^\circ$, $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = k$
 $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$, then the value of k is equal
 to
 (A) 4 (B) 6
 (C) 8 (D) 10

- Q14** If $A + B + C = 180^\circ$, prove that:

$$\sin^2 A + \sin^2 B - \sin^2 C = 2 \sin A \sin B \cos C.$$

- Q15** If $A + B + C = \pi$ and
 $\sin\left(A + \frac{C}{2}\right) = K \sin \frac{C}{2}$, then $\tan \frac{A}{2} \cdot \tan \frac{B}{2}$ is
 equal to
 (A) $\frac{K-1}{K+1}$
 (B) $\frac{K+1}{K-1}$
 (C) $\frac{K}{K+1}$
 (D) $\frac{K+1}{K}$



Answer Key

Q1 (B)

Q2 (D)

Q3 (A, B, C, D)

Q4 (C)

Q5 (A, C, D)

Q6 (A, B, C, D)

Q7 (B)

Q8 (C)

Q9 (B)

Q10 (B)

Q11 (C)

Q12 (D)

Q13 (C)

Q14 proof

Q15 (A)



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